

Artificial Intelligence and Machine Learning

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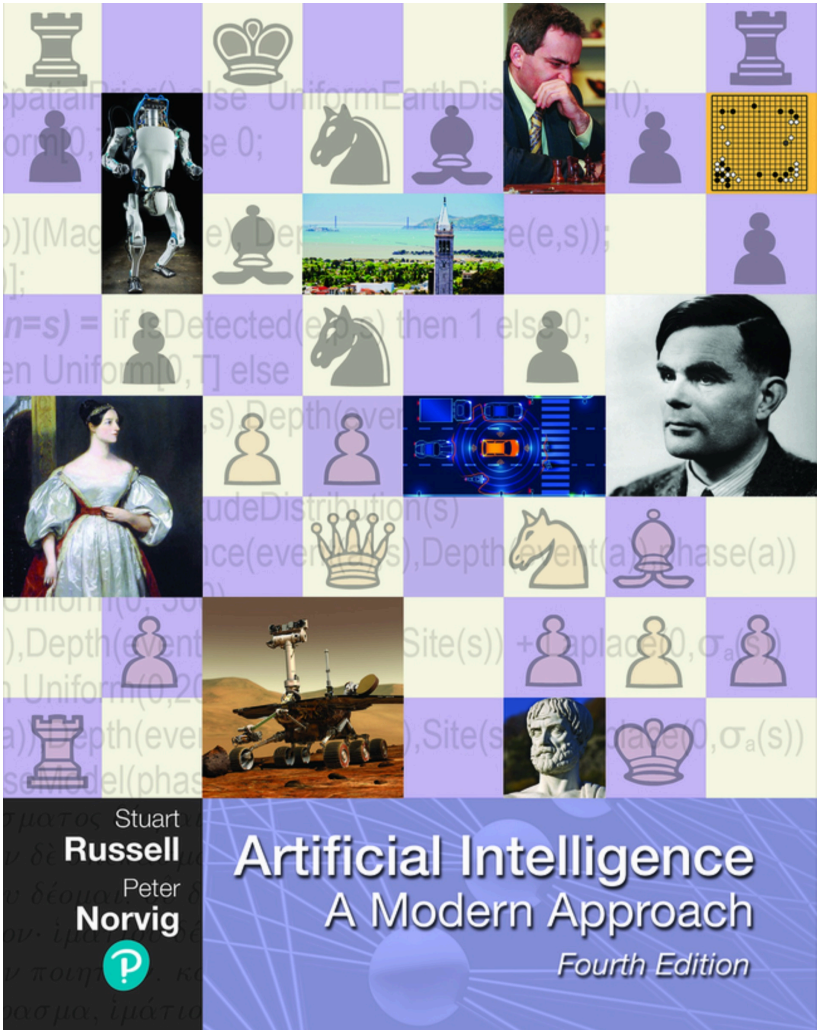
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The Machine Learning Context

Artificial Intelligence

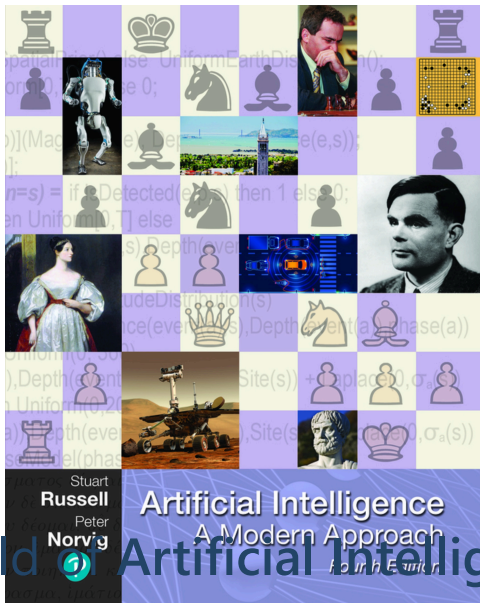
The field of **Artificial Intelligence** (AI) is concerned with *understanding and building intelligent entities*.

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Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th Edition)

Artificial Intelligence - Intelligent Entities Rows



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Artificial Intelligence - Intelligent Entities Rows and Columns



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Code Example

```
from sklearn.datasets import make_blobs
import matplotlib.pyplot as plt

</div>

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# Generate sample data

<div class="slide-content">
X, y = make_blobs(n_samples=100, centers=2, random_state=42)
</div>

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# Plot the data

<div class="slide-content">
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Sample ML Dataset")
plt.show()
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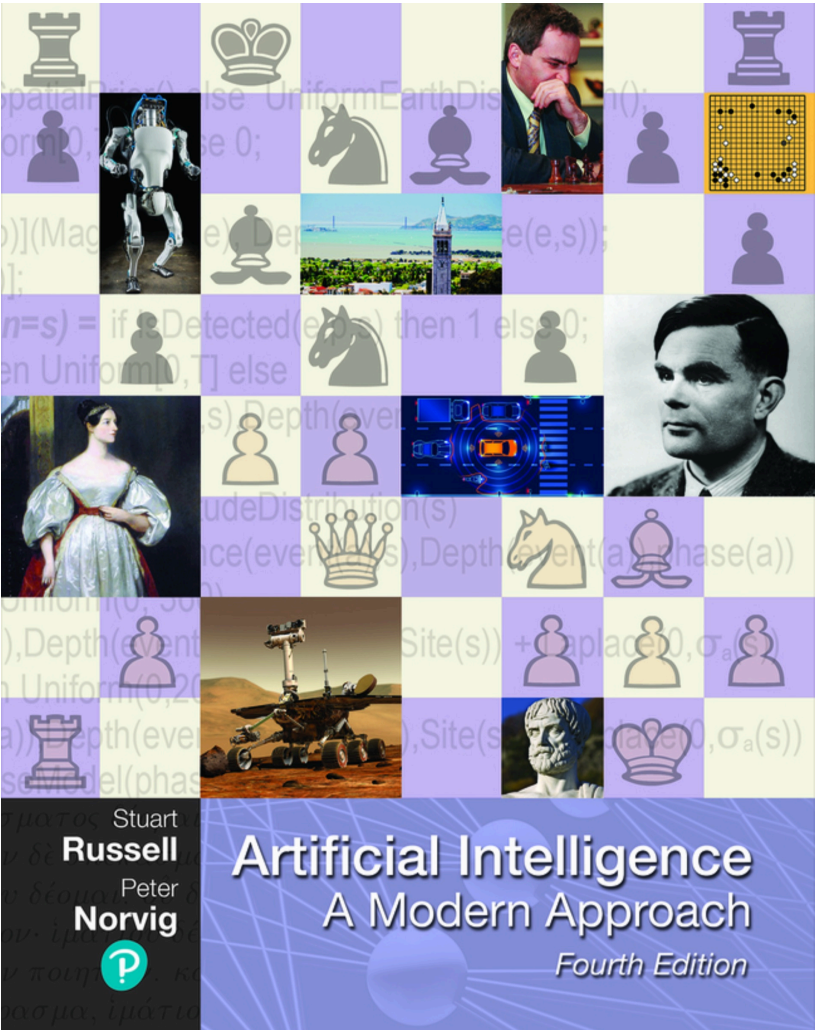
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Code Example in Columns



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Many Thanks!

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